Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-4 SWL Berm 2% - Test 2.experiment - 1000 (4).txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.250 Hz*
Freq. upper bound:	*5.000 Hz*

Freq. Domain Analysis

Overlap of subseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*32*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*38*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *33.1%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1238*	*0.04094*
Peak wave period T_P (s):	*1.939*	*1.939*
Energy wave period $T_{-1,0}$ (s):	*1.773*	*1.729*
Mean wave period $T_{0,1}$ (s):	*1.589*	*1.26*
Mean wave period $T_{0,2}$ (s):	*1.468*	*1.038*
Spectral width (Broadness):	*0.7708*	*0.8434*
Spectral width (Narrowness):	*0.4139*	*0.6877*

Time Domain Analysis

Overall amplitude reflection coefficient: *33.3%*

	Incident	**Reflected**
Number of waves:	994	1147
Mean wave height H_m (m):	*0.08055*	*0.02668*
Mean wave period T_m (s):	*1.673*	*1.449*
Sig. wave height H_s (m):	*0.1228*	*0.04366*
$T_{H_{1/3}}$ (s):	*1.888*	*1.838*
H_{max} (m):	*0.2132*	*0.08628*
T_{Hmax} (s):	*1.96*	*2.63*
$H_{1/10}$ (m):	*0.1506*	*0.05762*
$H_{1/50}$ (m):	*0.1799*	*0.07224*
$H_{1/100}$ (m)	*0.1936*	*0.07697*
$H_{1/250}$ (m):	*0.205*	*0.08143*
$H_{0.1\%}$ (m):	*Not enough data*	*0.08591*
$H_{1\%}$ (m):	*0.175*	*0.07285*
$H_{2\%}$ (m):	*0.161*	*0.06372*
$H_{10\%}$ (m)	*0.1312*	*0.04742*
Groupiness factor GF:	*1.049*	*1.229*
Skewness b1:	*0.7138*	*0.2971*
Kurtosis b2:	*3.301*	*3.799*

Department of Civil Engineering

Aalborg University

Project title	
Drawn by	
Date	*11. mrt 2026*
Remarks	
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Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-4 SWL Berm 2% - Test 2.experiment - 1000 (4).txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.242 Hz*
Freq. upper bound:	*4.844 Hz*

Freq. Domain Analysis

Overlap of subseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*34*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*40*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *35.4%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1377*	*0.04876*
Peak wave period T_P (s):	*2.065*	*2.065*
Energy wave period $T_{-1,0}$ (s):	*1.846*	*1.834*
Mean wave period $T_{0,1}$ (s):	*1.627*	*1.299*
Mean wave period $T_{0,2}$ (s):	*1.486*	*1.052*
Spectral width (Broadness):	*0.8011*	*0.8562*
Spectral width (Narrowness):	*0.447*	*0.7252*

Time Domain Analysis

Overall amplitude reflection coefficient: *35.5%*

	Incident	**Reflected**
Number of waves:	1030	1163
Mean wave height H_m (m):	*0.08875*	*0.03196*
Mean wave period T_m (s):	*1.738*	*1.54*
Sig. wave height H_s (m):	*0.1367*	*0.05157*
$T_{H_{1/3}}$ (s):	*1.996*	*1.973*
H_{max} (m):	*0.2239*	*0.0916*
T_{Hmax} (s):	*1.807*	*2.664*
$H_{1/10}$ (m):	*0.1682*	*0.06628*
$H_{1/50}$ (m):	*0.1972*	*0.07827*
$H_{1/100}$ (m)	*0.2087*	*0.08258*
$H_{1/250}$ (m):	*0.2215*	*0.08689*
$H_{0.1\%}$ (m):	*0.2239*	*0.0914*
$H_{1\%}$ (m):	*0.1924*	*0.07775*
$H_{2\%}$ (m):	*0.1795*	*0.07216*
$H_{10\%}$ (m)	*0.148*	*0.05607*
Groupiness factor GF:	*1.061*	*1.157*
Skewness b1:	*0.7894*	*0.1952*
Kurtosis b2:	*3.397*	*3.415*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-4 SWL Berm 2% - Test 2.experiment - 1200 (4).txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.227 Hz*
Freq. upper bound:	*4.531 Hz*

	Gauge 1	Gauge 2	Gauge 3
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Freq. Domain Analysis

Overlap of subseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*35*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*42*
Hilbert filter length:	*2048*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *37.8%*

	Incident	Reflected
Sig. wave height H_{m0} (m):	*0.1476*	*0.05578*
Peak wave period T_P (s):	*2.065*	*2.286*
Energy wave period $T_{-1,0}$ (s):	*1.897*	*1.921*
Mean wave period $T_{0,1}$ (s):	*1.646*	*1.33*
Mean wave period $T_{0,2}$ (s):	*1.488*	*1.064*
Spectral width (Broadness):	*0.8142*	*0.8596*
Spectral width (Narrowness):	*0.4723*	*0.7511*

Time Domain Analysis

Overall amplitude reflection coefficient: *37.9%*

	Incident	Reflected
Number of waves:	1020	1151
Mean wave height H_m (m):	*0.09824*	*0.03809*
Mean wave period T_m (s):	*1.809*	*1.603*
Sig. wave height H_s (m):	*0.1497*	*0.06061*
$T_{H_{1/3}}$ (s):	*2.089*	*2.022*
H_{max} (m):	*0.2267*	*0.1052*
T_{Hmax} (s):	*2.198*	*1.369*
$H_{1/10}$ (m):	*0.1805*	*0.07417*
$H_{1/50}$ (m):	*0.2027*	*0.08662*
$H_{1/100}$ (m)	*0.2118*	*0.09071*
$H_{1/250}$ (m):	*0.2237*	*0.09496*
$H_{0.1\%}$ (m):	*0.2267*	*0.1034*
$H_{1\%}$ (m):	*0.1987*	*0.08556*
$H_{2\%}$ (m):	*0.1912*	*0.08099*
$H_{10\%}$ (m)	*0.1618*	*0.06541*
Groupiness factor GF:	*1.051*	*1.086*
Skewness b1:	*0.8416*	*0.1228*
Kurtosis b2:	*3.384*	*3.193*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-4 SWL Berm 2% - Test 2.experiment - 1500 (4).txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.219 Hz*
Freq. upper bound:	*4.375 Hz*

	Gauge 1	Gauge 2	Gauge 3
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Freq. Domain Analysis

Overlap of subseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*37*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*44*
Hilbert filter length:	*2048*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *39.6%*

	Incident	Reflected
Sig. wave height H_{m0} (m):	*0.1597*	*0.06323*
Peak wave period T_P (s):	*2.286*	*2.286*
Energy wave period $T_{-1,0}$ (s):	*1.93*	*1.961*
Mean wave period $T_{0,1}$ (s):	*1.643*	*1.304*
Mean wave period $T_{0,2}$ (s):	*1.467*	*1.028*
Spectral width (Broadness):	*0.8317*	*0.8593*
Spectral width (Narrowness):	*0.5033*	*0.7805*

Time Domain Analysis

Overall amplitude reflection coefficient: *39.7%*

	Incident	Reflected
Number of waves:	1027	1140
Mean wave height H_m (m):	*0.1068*	*0.04382*
Mean wave period T_m (s):	*1.851*	*1.669*
Sig. wave height H_s (m):	*0.1628*	*0.0669*
$T_{H_{1/3}}$ (s):	*2.109*	*2.055*
H_{max} (m):	*0.2383*	*0.1275*
T_{Hmax} (s):	*2.204*	*1.376*
$H_{1/10}$ (m):	*0.1918*	*0.08159*
$H_{1/50}$ (m):	*0.2151*	*0.09542*
$H_{1/100}$ (m)	*0.2222*	*0.1019*
$H_{1/250}$ (m):	*0.2321*	*0.1101*
$H_{0.1\%}$ (m):	*0.2382*	*0.1246*
$H_{1\%}$ (m):	*0.2114*	*0.09356*
$H_{2\%}$ (m):	*0.2047*	*0.08712*
$H_{10\%}$ (m)	*0.1716*	*0.07186*
Groupiness factor GF:	*1.055*	*1.039*
Skewness b1:	*0.9212*	*0.04935*
Kurtosis b2:	*3.449*	*3.017*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-4 SWL Berm 2% - Test 2.experiment - 1000 (4).txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.211 Hz*
Freq. upper bound:	*4.219 Hz*

	Gauge 1	Gauge 2	Gauge 3
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Freq. Domain Analysis

Overlap of subseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*38*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*46*
Hilbert filter length:	*2048*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *40.9%*

	Incident	Reflected
Sig. wave height H_{m0} (m):	*0.1641*	*0.06717*
Peak wave period T_P (s):	*2.37*	*2.462*
Energy wave period $T_{-1,0}$ (s):	*1.955*	*2.035*
Mean wave period $T_{0,1}$ (s):	*1.638*	*1.328*
Mean wave period $T_{0,2}$ (s):	*1.452*	*1.035*
Spectral width (Broadness):	*0.8354*	*0.8639*
Spectral width (Narrowness):	*0.5225*	*0.8051*

Time Domain Analysis

Overall amplitude reflection coefficient: *41%*

	Incident	Reflected
Number of waves:	1039	1095
Mean wave height H_m (m):	*0.1119*	*0.04885*
Mean wave period T_m (s):	*1.893*	*1.796*
Sig. wave height H_s (m):	*0.168*	*0.07136*
$T_{H_{1/3}}$ (s):	*2.188*	*2.188*
H_{max} (m):	*0.2379*	*0.1225*
T_{Hmax} (s):	*2.157*	*1.834*
$H_{1/10}$ (m):	*0.194*	*0.08534*
$H_{1/50}$ (m):	*0.2156*	*0.09985*
$H_{1/100}$ (m)	*0.2229*	*0.1056*
$H_{1/250}$ (m):	*0.2299*	*0.1114*
$H_{0.1\%}$ (m):	*0.2375*	*0.1216*
$H_{1\%}$ (m):	*0.2131*	*0.09766*
$H_{2\%}$ (m):	*0.2058*	*0.09175*
$H_{10\%}$ (m)	*0.1776*	*0.07522*
Groupiness factor GF:	*1.038*	*0.9781*
Skewness b1:	*0.9472*	*-0.02988*
Kurtosis b2:	*3.399*	*2.848*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-4 SWL Berm 2% - Test 2.experiment - 1500 (4).txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.203 Hz*
Freq. upper bound:	*4.063 Hz*

	Gauge 1	Gauge 2	Gauge 3
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Freq. Domain Analysis

50% overlap of subseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*39*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*47*
Hilbert filter length:	*2048*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *42%*

	Incident	Reflected
Sig. wave height H_{m0} (m):	*0.1717*	*0.07209*
Peak wave period T_P (s):	*2.462*	*2.462*
Energy wave period $T_{-1,0}$ (s):	*1.98*	*2.082*
Mean wave period $T_{0,1}$ (s):	*1.64*	*1.335*
Mean wave period $T_{0,2}$ (s):	*1.446*	*1.035*
Spectral width (Broadness):	*0.8387*	*0.86*
Spectral width (Narrowness):	*0.5354*	*0.8149*

Time Domain Analysis

Overall amplitude reflection coefficient: *42.2%*

	Incident	Reflected
Number of waves:	1023	1147
Mean wave height H_m (m):	*0.1192*	*0.05145*
Mean wave period T_m (s):	*1.968*	*1.755*
Sig. wave height H_s (m):	*0.1766*	*0.07528*
$T_{H_{1/3}}$ (s):	*2.231*	*2.171*
H_{max} (m):	*0.25*	*0.1204*
T_{Hmax} (s):	*1.284*	*1.368*
$H_{1/10}$ (m):	*0.2033*	*0.08919*
$H_{1/50}$ (m):	*0.2241*	*0.102*
$H_{1/100}$ (m)	*0.231*	*0.1066*
$H_{1/250}$ (m):	*0.2391*	*0.1134*
$H_{0.1\%}$ (m):	*0.2497*	*0.1203*
$H_{1\%}$ (m):	*0.2229*	*0.09956*
$H_{2\%}$ (m):	*0.2142*	*0.09626*
$H_{10\%}$ (m)	*0.1851*	*0.07981*
Groupiness factor GF:	*1.048*	*0.9541*
Skewness b1:	*0.999*	*-0.07541*
Kurtosis b2:	*3.479*	*2.79*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-4 SWL Berm 2% - Test 2.experiment - 60% (4).txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.336 Hz*
Freq. upper bound:	*6.719 Hz*

Freq. Domain Analysis

Overlap of subseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*25*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*29*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *22.7%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.0821*	*0.01863*
Peak wave period T_P (s):	*1.488*	*1.561*
Energy wave period $T_{-1,0}$ (s):	*1.44*	*1.344*
Mean wave period $T_{0,1}$ (s):	*1.365*	*1.081*
Mean wave period $T_{0,2}$ (s):	*1.316*	*0.9434*
Spectral width (Broadness):	*0.6017*	*0.7979*
Spectral width (Narrowness):	*0.2758*	*0.5599*

Time Domain Analysis

Overall amplitude reflection coefficient: *22.6%*

	Incident	**Reflected**
Number of waves:	968	1235
Mean wave height H_m (m):	*0.05236*	*0.01145*
Mean wave period T_m (s):	*1.354*	*1.061*
Sig. wave height H_s (m):	*0.0804*	*0.01832*
$T_{H_{1/3}}$ (s):	*1.469*	*1.352*
H_{max} (m):	*0.1217*	*0.03414*
T_{Hmax} (s):	*1.464*	*1.454*
$H_{1/10}$ (m):	*0.09578*	*0.02366*
$H_{1/50}$ (m):	*0.1106*	*0.02833*
$H_{1/100}$ (m)	*0.1156*	*0.03011*
$H_{1/250}$ (m):	*0.118*	*0.03213*
$H_{0.1\%}$ (m):	*Not enough data*	*0.03379*
$H_{1\%}$ (m):	*0.1107*	*0.02744*
$H_{2\%}$ (m):	*0.1037*	*0.02582*
$H_{10\%}$ (m)	*0.08515*	*0.02*
Groupiness factor GF:	*0.9286*	*1.102*
Skewness b1:	*0.3155*	*0.1249*
Kurtosis b2:	*2.816*	*3.341*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-4 SWL Berm 2% - Test 2.experiment - 80% (4).txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.281 Hz*
Freq. upper bound:	*5.625 Hz*

Freq. Domain Analysis

Overlap of subseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*28*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*34*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *27.9%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1073*	*0.02998*
Peak wave period T_P (s):	*1.778*	*1.882*
Energy wave period $T_{-1,0}$ (s):	*1.633*	*1.559*
Mean wave period $T_{0,1}$ (s):	*1.51*	*1.179*
Mean wave period $T_{0,2}$ (s):	*1.427*	*0.9908*
Spectral width (Broadness):	*0.7172*	*0.8314*
Spectral width (Narrowness):	*0.3471*	*0.6445*

Time Domain Analysis

Overall amplitude reflection coefficient: *27.9%*

	Incident	**Reflected**
Number of waves:	962	1185
Mean wave height H_m (m):	*0.06891*	*0.01888*
Mean wave period T_m (s):	*1.546*	*1.255*
Sig. wave height H_s (m):	*0.1046*	*0.03071*
$T_{H_{1/3}}$ (s):	*1.698*	*1.63*
H_{max} (m):	*0.1627*	*0.06964*
T_{Hmax} (s):	*1.456*	*1.427*
$H_{1/10}$ (m):	*0.1281*	*0.03986*
$H_{1/50}$ (m):	*0.1502*	*0.05196*
$H_{1/100}$ (m)	*0.1556*	*0.05797*
$H_{1/250}$ (m):	*0.1598*	*0.066*
$H_{0.1\%}$ (m):	*Not enough data*	*0.06942*
$H_{1\%}$ (m):	*0.1493*	*0.04994*
$H_{2\%}$ (m):	*0.1427*	*0.04385*
$H_{10\%}$ (m)	*0.1104*	*0.03284*
Groupiness factor GF:	*0.9777*	*1.219*
Skewness b1:	*0.5058*	*0.3307*
Kurtosis b2:	*2.967*	*3.83*